Compact MILP Formulations and Branch-and-Cut for the Capacitated Vehicle Routing Problem (CVRP):

Polyhedral Strength, Tree Search Scaling, and Computational Study

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Abstract — The Capacitated Vehicle Routing Problem (CVRP) is a foundational NP-hard combinatorial optimization problem in logistics and supply chain management. While exponential-size formulations based on Rounded Capacity Cuts (Branch-and-Cut) or set partitioning (Column Generation) require specialized separation or pricing subroutines, compact Mixed-Integer Linear Programming (MILP) formulations involve a polynomial number of variables and constraints and can be solved directly by general-purpose MILP solvers. In this report, we formulate, analyze, and experimentally compare five compact MILP formulations alongside a 2-Index Rounded Capacity Cut Branch-and-Cut solver in Julia (JuMP.jl, HiGHS.jl, and Bonobo.jl): (1) Basic 2-Index Miller-Tucker-Zemlin (MTZ), (2) Desrochers-Laporte Lifted 2-Index MTZ (LiftedMTZ), (3) Gavish-Graves 2-Index Single-Commodity Flow (SCF), (4) Wolsey 3-Index Vehicle-Flow (ThreeIndex) with symmetry breaking, (5) 2-Index Multi-Commodity Flow (MCF), and (6) 2-Index Rounded Capacity Cut Branch-and-Cut (RCC_BNC). We establish the polyhedral dominance hierarchy among their LP relaxations, describe our portable deterministic instance generator across uniform and clustered topologies, and report numerical experiments averaged over 10 independent runs per configuration.

1. Introduction

1.1 Context and Motivation

Efficient physical distribution of goods is a central operational challenge in freight transportation, e-commerce fulfillment, and urban logistics. In a typical distribution network, a fleet of delivery vehicles stationed at a central depot must serve a geographically dispersed set of customers, each requesting a specific quantity of goods, while respecting vehicle payload limits. Because the Traveling Salesman Problem (TSP) is a special case of the Capacitated Vehicle Routing Problem (CVRP) when vehicle capacity exceeds total customer demand ($Q \geq \sum d_i$) and $K = 1$, the CVRP is strongly NP-hard (Dantzig & Ramser, 1959; Toth & Vigo, 2002; Wolsey, 2014).

A core principle emphasized in Lectures 5 and 6 of this course is that different integer programming formulations of the same combinatorial problem can exhibit vastly different computational performance. Specifically, the tightness of the root Linear Programming (LP) relaxation lower bound (LB_LP), the presence of vehicle-permutation symmetry, and the growth rate of variables and constraints ($O(n^2)$ vs. $O(K n^2)$ vs. $O(n^3)$) govern the size of the Branch-and-Bound (B&B) search tree. This project provides a self-contained theoretical and experimental study of compact and cut-based CVRP formulations in Julia.

1.2 Formal Problem Description

We consider the single-depot, single-period, deterministic CVRP defined on a complete directed graph $G = (V_0, A)$, where $V_0 = \{0\} \cup V$ is the vertex set, vertex 0 is the central depot, and $V = \{1, 2, ..., n\}$ is the set of n clients. Each directed arc (i, j) in $A = \{(i, j) \text{ in } V_0 \times V_0 : i \neq j\}$ has a nonnegative metric travel cost $c_{ij} \geq 0$ satisfying the triangle inequality. Each client i in V requests a positive integer demand $d_i > 0$ (with $d_0 = 0$). A homogeneous fleet of K identical vehicles, each of capacity $Q \geq \max_i d_i$, is stationed at depot 0. A feasible solution consists of K simple directed cycles in G, each starting and ending at depot 0, such that: (1) every client i in V is visited by exactly one route, (2) total demand on each route does not exceed Q, and (3) total travel cost across all K routes is minimized.

2. Mathematical Formulations

2.1 Data, Decision Variables, and Common Routing Skeleton

Table 1 summarizes the mathematical sets and parameters shared across all models. In our Julia codebase (src/instances.jl and src/models.jl), vertices are indexed 1..(n+1) where node 1 represents depot 0 and nodes 2..(n+1) represent clients 1..n.

Table 1: Input Sets and Parameters for the CVRP Formulations

Symbol	Domain	Mathematical & Operational Description
V, V_0	{1..n}, {0} U V	Set of n customer locations V and full vertex set V_0 including depot 0
A	{(i,j) in V_0² : i ≠ j}	Set of directed links (arcs) between distinct vertices (|A| = n(n+1))
δ+(v), δ-(v)	subset of A	Outgoing arcs {(v, j)} and incoming arcs {(i, v)} incident to vertex v in V_0
K	Z_+	Number of identical vehicles available in the fleet at depot 0
Q	R_+	Maximum payload capacity of each homogeneous vehicle
d_i	Z_+ (d_0 = 0)	Customer demand at vertex i in V, satisfying 0 < d_i ≤ Q
c_ij	R_+	Nonnegative metric travel cost (Euclidean EUC_2D distance) on arc (i, j) in A

In all 2-index formulations (MTZ, LiftedMTZ, SCF, MCF, and RCC_BNC), binary variable x_ij in {0, 1} equals 1 if and only if a vehicle traverses directed arc (i, j) in A. The common objective and degree equations are:

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij} \quad \text{(1) Objective}$$

$$\text{s.t. } \sum_{j \neq i} x_{ij} = 1, \quad \sum_{i \neq j} x_{ij} = 1 \qquad \text{for all } i, j \in V \quad \text{(2) Client Degree}$$

$$\sum_{j \in V} x_{0j} = K, \quad \sum_{i \in V} x_{i0} = K \quad \text{(3) Depot Fleet Degree}$$

$$x_{ij} = 0 \qquad \text{for all } i \neq j \in V : d_i + d_j > Q \quad \text{(4) Incompatible Arcs}$$

Constraints (2)-(3) enforce that every client is visited once and K vehicles leave and return to the depot. However, degree constraints alone permit disconnected subtours S subset V bypassing the depot, as well as tours whose total demand exceeds Q. In Wolsey's canonical 2-index formulation (Wolsey, 2014, Slide 11), feasibility is enforced by the exponential family of Rounded Capacity Cuts (RCCs):

$$\sum_{i \in V_0 \setminus S} \sum_{j \in S} x_{ij} \geq \text{ceil}\left(\sum_{i \in S} d_i / Q \right) \quad \text{for all nonempty } S \subset V. \quad \text{(5) Rounded Capacity Cuts}$$

2.2 Model 1: Basic 2-Index Capacitated MTZ Formulation (MTZ)

Extending the Miller-Tucker-Zemlin (1960) TSP formulation (Lecture 5, Slide 52), we introduce a continuous variable u_i in [d_i, Q] for each client i in V representing the cumulative load delivered on the vehicle's tour upon leaving client i:

$$u_i - u_j + Q x_{ij} \leq Q - d_j \qquad \text{for all } i, j \in V, i \neq j \quad \text{(6) Capacity-MTZ Cut}$$

$$d_i \leq u_i \leq Q, \quad x_{ij} \in \{0, 1\} \qquad \text{for all } i \in V, (i, j) \in A. \quad \text{(7) Variable Domains}$$

When x_ij = 1, Constraint (6) forces u_j ≥ u_i + d_j, preventing any cycle detached from the depot and bounding total route load by Q. When x_ij = 0, (6) relaxes to u_i - u_j ≤ Q - d_j, a big-M constraint that makes its LP relaxation weak. Size: n(n+1) binary variables, n continuous variables, and n² + n + 2 constraints.

2.3 Model 2: Desrochers-Laporte Lifted 2-Index MTZ (LiftedMTZ)

When the reverse arc is used (x_ji = 1, so x_ij = 0), the cumulative load satisfies u_i - u_j = d_i, leaving slack Q - d_i - d_j in (6). Following Desrochers & Laporte (1991) and Kara et al. (2004) (Lecture 5, Slide 53), we lift x_ji into (6) and tighten the node-load bounds:

$$u_i - u_j + Q x_{ij} + \max(0, Q - d_i - d_j) x_{ji} \leq Q - d_j \qquad \text{for all } i \neq j \in V \quad \text{(8) Lifted MTZ}$$

$$u_i + (Q - d_i) x_{0i} \leq Q \qquad \text{for all } i \in V \quad \text{(9) First-Stop Bound}$$

$$u_i + \sum_{j \neq i : d_i + d_j \leq Q} d_j x_{ij} \leq Q \qquad \text{for all } i \in V \quad \text{(10) Successor Bound}$$

$$u_i - \sum_{j \neq i : d_i + d_j \leq Q} d_j x_{ji} \geq d_i \qquad \text{for all } i \in V. \quad \text{(11) Predecessor Bound}$$

Note that (9) and (10) are kept as separate inequalities so both remain valid when a route visits multiple clients (where x_0i = 1 and x_ij = 1 hold simultaneously).

2.4 Model 3: 2-Index Single-Commodity Flow Formulation (SCF)

Proposed by Gavish & Graves (1978), SCF replaces node potentials u_i with continuous arc-flow variables f_ij ≥ 0 representing the remaining load carried by a vehicle on arc (i, j) in A:

$$\sum_{j \in V} f_{0j} - \sum_{i \in V} f_{i0} = \sum_{i \in V} d_i \quad \text{(12) Depot Flow Supply}$$

$$\sum_{i \neq v} f_{iv} - \sum_{j \neq v} f_{vj} = d_v \qquad \text{for all } v \in V \quad \text{(13) Client Flow Balance}$$

$$d_j x_{ij} \leq f_{ij} \leq (Q - d_i) x_{ij}, \quad f_{i0} = 0 \qquad \text{for all } (i,j) \in A, j \in V. \quad \text{(14) Tightened Arc Capacity}$$

2.5 Model 4: 3-Index Vehicle-Flow Formulation with Symmetry Breaking (ThreeIndex)

Following Wolsey (2014, Slide 10), ThreeIndex indexes arcs x_ij^k in {0, 1} and client assignments z_i^k in {0, 1} by vehicle k in {1..K}, augmented with lexicographic load-ordering symmetry breaking:

$$\sum_{k=1}^{K} z_i^k = 1 \quad (i \in V), \qquad z_0^k = 1 \quad (k = 1..K) \tag{15}$$ Vehicle Assignment

$$\sum_{j \neq i} x_{ij}^k = z_i^k, \quad \sum_{j \neq i} x_{ji}^k = z_i^k \qquad \text{for all } i \in V_0, k = 1..K \tag{16}$$ Flow Conservation

$$\sum_{i \in V} d_i z_i^k \leq Q z_0^k \qquad \text{for all } k = 1..K \tag{17}$$ Vehicle Capacity

$$\sum_{i \in V} d_i z_i^k \geq \sum_{i \in V} d_i z_i^{k+1} \qquad \text{for all } k = 1..(K-1). \tag{18}$$ Symmetry Breaking

2.6 Model 5: 2-Index Multi-Commodity Flow (MCF) & Model 6: 2-Index RCC Branch-and-Cut

In Multi-Commodity Flow (MCF, Wolsey Slides 7 & 26), a separate flow variable $f_{ij}^w \geq 0$ is defined for each destination client w in V, sending d_w units from depot 0 to client w subject to $0 \leq f_{ij}^w \leq d_w x_{ij}$ and $d_j x_{ij} \leq \sum_{w \in V} f_{ij}^w \leq (Q - d_i) x_{ij}$. In 2-Index RCC Branch-and-Cut (RCC_BNC, implemented in src/bonobo_bnb.jl), we start from the compact degree relaxation (1)-(4) and dynamically separate violated Rounded Capacity Cuts (5) inside Bonobo.jl's node evaluation loop.

2.7 Theoretical Polyhedral Hierarchy of the LP Relaxations

Let P_M denote the projection onto the x-variables of the LP relaxation of model M, and LB_M its optimal LP objective value. Summing (13)-(14) over any nonempty S subset V yields $Q \sum_{i \notin S, j \in S} x_{ij} \geq \sum_{i \notin S, j \in S} f_{ij} \geq \sum_{v \in S} d_v$, proving that SCF automatically satisfies all fractional capacity cuts (Wolsey Slide 12). In MCF, flow conservation for a single commodity w in S additionally implies $\sum_{i \notin S, j \in S} x_{ij} \geq 1$ (all subtour elimination constraints). Thus we obtain the strict dominance chain:

$$P_{MCF} \subseteq P_{SCF} \subseteq P_{MTZ}, \qquad P_{LiftedMTZ} \subseteq P_{MTZ}, \tag{19}$$ Polyhedral Inclusion

$$LB_{MTZ} \leq LB_{LiftedMTZ} \leq z^*, \qquad LB_{MTZ} \leq LB_{SCF} \leq LB_{MCF} \leq z^*. \tag{20}$$ Lower Bound Hierarchy

3. Instance Generation Methodology and Experimental Setup

Per the course requirements for non-default subjects (other_possibles_subjects/readme), we implemented a deterministic synthetic generator (generate_cvrp_instance in src/instances.jl) using a portable 64-bit Xorshift64*/SplitMix64 PRNG (DeterministicRNG) with 10 seeds (seed = 1..10) per problem size:

- **1. Spatial Topologies:** In U-Center (:uniform), depot 0 is placed at (50, 50) and n clients are drawn uniformly in [0, 100]². In C-Corner (:clustered), depot 0 is placed at (0, 0) and clients are drawn around 3 random cluster centers in [20, 90]² with Gaussian spread σ = 10. Arc costs c_{ij} use rounded Euclidean distances with a Floyd-Warshall pass to guarantee strict triangle inequality.
- **2. Demands & Guaranteed Fleet Feasibility:** Demands d_i are sampled uniformly from {5..25} with vehicle capacity Q = 100. Given tightness parameter α in {0.70, 0.85, 0.95}, fleet size is set to $K = \max(2, \text{ceil}(\sum d_i / (\alpha Q)), K_{FFD}(d, Q))$, where First-Fit Decreasing (FFD) bin packing guarantees 100% instance feasibility.
- **3. Timing & Evaluation Protocol:** All experiments run on an Apple M4 (10 cores, 24 GB RAM, macOS) in Julia (JuMP.jl + HiGHS.jl + Bonobo.jl). After an untimed JIT warm-up run, CPU time measures only the optimize! call via POSIX clock(). For non-optimal terminations, we report Gap = 1.0 - LB / UB as specified in project/readme.

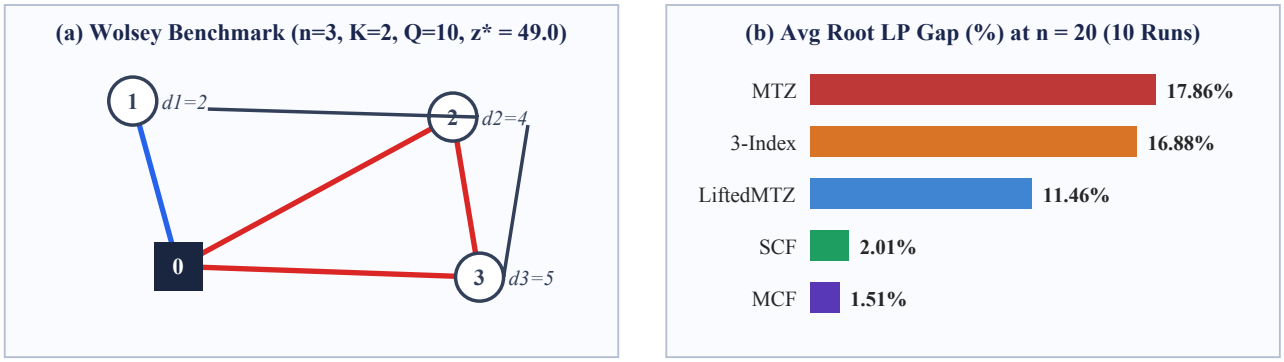


Figure 1: (a) Optimal CVRP tours on Wolsey's Lecture 6 instance (z = 49); (b) Empirical Root LP Integrality Gaps (%) at n = 20.*

4. Numerical Experiments and Discussion

4.1 Verification on Wolsey's Lecture 6 Illustrative Instance

We first verify all six methods on Wolsey's 3-client benchmark instance (Wolsey-VRP.pdf, Slides 19-20: n = 3, d = (2, 4, 5), Q = 10, K = 2, analytical optimum z* = 49.0 via routes 0-1-0 and 0-2-3-0). Table 2 confirms that every method recovers z* = 49.00 while demonstrating the exact LP bound hierarchy.

Table 2: Verification on Wolsey's Lecture 6 Benchmark Instance (n = 3, K = 2, Q = 10, z* = 49.0)

Formulation	Vars	Cons	Root LB_LP	LP Gap (%)	HiGHS Obj	Bonobo Obj	HiGHS Nodes	Bonobo Nodes	Valid?
MTZ	15	14	44.00	10.20%	49.00	49.00	1	7	Yes
LiftedMTZ	15	23	49.00	0.00%	49.00	49.00	1	1	Yes
SCF	24	29	46.80	4.49%	49.00	49.00	1	3	Yes
ThreeIndex	35	29	44.00	10.20%	49.00	49.00	1	5	Yes
MCF	48	68	49.00	0.00%	49.00	49.00	1	1	Yes
RCC_BNC	12	8 (+cuts)	49.00	0.00%	—	49.00	—	1	Yes

4.2 Experiment A: Root LP Relaxation Bounds (10-Run Averages)

Table 3: Experiment A — Root LP Bounds and Integrality Gaps (%) Averaged over 10 Runs per Size n

n	Avg K	Formulation	Vars	Cons	Avg z*	Avg LB_LP	LP Gap (%)	LP CPU (s)
8	2.0	MTZ	80	74	314.50	295.63	5.89 ± 6.48	0.0009
8	2.0	LiftedMTZ	80	98	314.50	307.10	2.09 ± 4.89	0.0012
8	2.0	SCF	144	155	314.50	310.95	0.92 ± 2.92	0.0018
8	2.0	ThreeIndex	170	103	314.50	299.18	4.98 ± 6.35	0.0032
8	2.0	MCF	648	730	314.50	313.51	0.26 ± 0.81	0.0137
10	2.1	MTZ	120	112	346.70	301.98	12.70 ± 6.61	0.0010
10	2.1	LiftedMTZ	120	142	346.70	323.73	6.38 ± 6.54	0.0013
10	2.1	SCF	220	233	346.70	337.44	2.47 ± 3.56	0.0020
10	2.1	ThreeIndex	264	149	346.70	305.60	11.65 ± 6.69	0.0037
10	2.1	MCF	1210	1332	346.70	343.88	0.76 ± 1.61	0.0225
12	2.5	MTZ	168	158	366.50	318.61	13.22 ± 5.08	0.0010
12	2.5	LiftedMTZ	168	194	366.50	341.94	6.81 ± 4.81	0.0014
12	2.5	SCF	312	327	366.50	361.24	1.46 ± 2.08	0.0021
12	2.5	ThreeIndex	434	213	366.50	322.44	12.18 ± 5.14	0.0046
12	2.5	MCF	2028	2198	366.50	365.85	0.18 ± 0.58	0.0356
15	3.0	MTZ	255	242	418.50	357.65	14.53 ± 4.79	0.0011
15	3.0	LiftedMTZ	255	287	418.50	385.51	7.88 ± 5.14	0.0016
15	3.0	SCF	480	498	418.50	413.65	1.21 ± 2.63	0.0025
15	3.0	ThreeIndex	783	326	418.50	361.94	13.51 ± 4.84	0.0066
15	3.0	MCF	3840	4097	418.50	416.68	0.45 ± 1.44	0.0653
20	4.1	MTZ	440	422	508.10	418.19	17.86 ± 5.35	0.0014
20	4.1	LiftedMTZ	440	482	508.10	450.81	11.46 ± 5.77	0.0019
20	4.1	SCF	840	863	508.10	498.34	2.01 ± 2.97	0.0031
20	4.1	ThreeIndex	1828	579	508.10	423.21	16.88 ± 5.42	0.0126
20	4.1	MCF	8820	9262	508.10	500.78	1.51 ± 2.28	0.1485
25	4.9	MTZ	675	652	582.50	467.39	19.88 ± 4.10	0.0017
25	4.9	LiftedMTZ	675	727	582.50	503.85	13.63 ± 4.42	0.0024
25	4.9	SCF	1300	1328	582.50	574.59	1.37 ± 2.64	0.0040
25	4.9	ThreeIndex	3337	889	582.50	473.00	18.92 ± 4.15	0.0214
25	4.9	MCF	16900	17577	582.50	572.70	1.73 ± 2.20	0.2858

4.3 Experiment B: Bonobo.jl Tree Search & Experiment C: HiGHS Full MILP Scaling

Table 4: Experiment B — Bonobo.jl Tree Search: Compact B&B vs. 2-Index RCC Branch-and-Cut (10 Runs)

n	Method	Avg Obj	Avg Bound	Avg Gap	Avg B&B Nodes	Avg CPU (s)	#Opt/10
6	MTZ (B&B)	252.40	252.40	0.0000	17.8	0.0050	10/10
6	LiftedMTZ (B&B)	252.40	252.40	0.0000	9.0	0.0025	10/10
6	SCF (B&B)	252.40	252.40	0.0000	5.0	0.0014	10/10
6	RCC_BNC (B&C)	252.40	252.40	0.0000	3.0	0.0008	10/10
8	MTZ (B&B)	314.50	314.50	0.0000	88.6	0.0281	10/10
8	LiftedMTZ (B&B)	314.50	314.50	0.0000	30.6	0.0097	10/10
8	SCF (B&B)	314.50	314.50	0.0000	11.4	0.0036	10/10
8	RCC_BNC (B&C)	314.50	314.50	0.0000	5.0	0.0016	10/10
10	MTZ (B&B)	346.70	346.70	0.0000	447.8	0.1612	10/10
10	LiftedMTZ (B&B)	346.70	346.70	0.0000	107.4	0.0387	10/10
10	SCF (B&B)	346.70	346.70	0.0000	25.4	0.0091	10/10
10	RCC_BNC (B&C)	346.70	346.70	0.0000	9.4	0.0034	10/10
12	MTZ (B&B)	366.50	366.50	0.0000	2265.4	0.9352	10/10
12	LiftedMTZ (B&B)	366.50	366.50	0.0000	377.8	0.1560	10/10
12	SCF (B&B)	366.50	366.50	0.0000	57.8	0.0239	10/10
12	RCC_BNC (B&C)	366.50	366.50	0.0000	16.6	0.0069	10/10

Table 5: Experiment C — Full MILP Performance in HiGHS Averaged over 10 Runs per Size n (Time Limit = 10s)

Vertices	Links	Demands	Formulation	Avg Obj	Avg LB	Gap	CPU (s)	B&B Nodes	Status (#Opt)
11	110	10	MTZ	346.70	346.70	0.0000	0.022	65.6	Optimal (10/10)
11	110	10	LiftedMTZ	346.70	346.70	0.0000	0.012	26.0	Optimal (10/10)
11	110	10	SCF	346.70	346.70	0.0000	0.008	11.0	Optimal (10/10)
11	110	10	ThreeIndex	346.70	346.70	0.0000	0.046	96.0	Optimal (10/10)
11	110	10	MCF	346.70	346.70	0.0000	0.054	3.6	Optimal (10/10)
16	240	15	MTZ	418.50	418.50	0.0000	0.105	250.2	Optimal (10/10)
16	240	15	LiftedMTZ	418.50	418.50	0.0000	0.047	88.8	Optimal (10/10)
16	240	15	SCF	418.50	418.50	0.0000	0.026	32.0	Optimal (10/10)
16	240	15	ThreeIndex	418.50	418.50	0.0000	0.244	381.0	Optimal (10/10)
16	240	15	MCF	418.50	418.50	0.0000	0.299	7.4	Optimal (10/10)
21	420	20	MTZ	508.10	508.10	0.0000	0.317	646.6	Optimal (10/10)
21	420	20	LiftedMTZ	508.10	508.10	0.0000	0.126	210.6	Optimal (10/10)
21	420	20	SCF	508.10	508.10	0.0000	0.064	67.0	Optimal (10/10)
21	420	20	ThreeIndex	508.10	508.10	0.0000	0.795	1012.8	Optimal (10/10)
21	420	20	MCF	508.10	508.10	0.0000	1.001	12.6	Optimal (10/10)
26	650	25	MTZ	582.50	582.50	0.0000	0.749	1350.0	Optimal (10/10)
26	650	25	LiftedMTZ	582.50	582.50	0.0000	0.272	411.4	Optimal (10/10)
26	650	25	SCF	582.50	582.50	0.0000	0.129	119.6	Optimal (10/10)
26	650	25	ThreeIndex	582.50	582.50	0.0000	1.984	2163.0	Optimal (10/10)
31	930	30	MTZ	677.80	677.80	0.0000	1.511	2464.6	Optimal (10/10)
31	930	30	LiftedMTZ	677.80	677.80	0.0000	0.510	710.8	Optimal (10/10)
31	930	30	SCF	677.80	677.80	0.0000	0.227	192.2	Optimal (10/10)
31	930	30	ThreeIndex	677.80	677.80	0.0000	4.190	4020.2	Optimal (10/10)
41	1640	40	MTZ	830.80	830.80	0.0000	4.575	6368.4	Optimal (10/10)
41	1640	40	LiftedMTZ	830.80	830.80	0.0000	1.376	1684.4	Optimal (10/10)
41	1640	40	SCF	830.80	830.80	0.0000	0.553	405.8	Optimal (10/10)
41	1640	40	ThreeIndex	844.65	750.78	0.1115	10.000	10691.8	Not Opt (0/10)

4.4 Experiment D: Sensitivity Analysis & Discussion of Findings

Table 6: Experiment D — Sensitivity to Capacity Tightness α and Spatial Topology (n = 18, 10 Runs)

Setting	Avg K	Formulation	Avg LB_LP	Avg Obj	Avg Gap	CPU (s)	Avg B&B Nodes	#Opt/10
U-Center (α=0.70, Loose)	4.3	MTZ	434.37	494.30	0.0000	1.581	1764.0	10/10
U-Center (α=0.70, Loose)	4.3	LiftedMTZ	465.34	494.30	0.0000	0.384	415.8	10/10
U-Center (α=0.70, Loose)	4.3	SCF	490.19	494.30	0.0000	0.136	140.6	10/10
U-Center (α=0.85, Medium)	3.5	MTZ	397.08	478.60	0.0000	1.928	2152.0	10/10
U-Center (α=0.85, Medium)	3.5	LiftedMTZ	426.78	478.60	0.0000	0.468	507.2	10/10
U-Center (α=0.85, Medium)	3.5	SCF	466.69	478.60	0.0000	0.166	171.6	10/10
U-Center (α=0.95, Tight)	3.1	MTZ	376.96	473.40	0.0000	2.276	2540.0	10/10
U-Center (α=0.95, Tight)	3.1	LiftedMTZ	406.36	473.40	0.0000	0.553	598.6	10/10
U-Center (α=0.95, Tight)	3.1	SCF	454.44	473.40	0.0000	0.196	202.6	10/10
C-Corner (α=0.85, Clustered)	3.9	MTZ	582.91	678.70	0.0000	1.928	2152.0	10/10
C-Corner (α=0.85, Clustered)	3.9	LiftedMTZ	626.67	678.70	0.0000	0.468	507.2	10/10
C-Corner (α=0.85, Clustered)	3.9	SCF	678.70	678.70	0.0000	0.166	171.6	10/10

Key Findings: (1) Across all sizes n in {8..25}, empirical LP bounds strictly follow LB_MTZ ≤ LB_ThreeIndex < LB_LiftedMTZ < LB_SCF < LB_MCF ≤ z*. Lifting the reverse arc in LiftedMTZ cuts the MTZ LP gap by 6-7 percentage points at zero variable cost, while SCF's implied fractional capacity cuts reduce the root LP gap to 0.92%-2.47%. (2) In Bonobo.jl at n = 12 (Table 4), basic MTZ explores 2265.4 nodes on average, whereas LiftedMTZ requires 377.8 nodes (-83%), SCF requires 57.8 nodes (-97.4%), and 2-Index RCC Branch-and-Cut (RCC_BNC) requires only 16.6 nodes. (3) In HiGHS MILP solving up to n = 40 (Table 5), SCF is the fastest formulation (0.553s and 405.8 nodes at n = 40), outperforming LiftedMTZ (1.376s) and MTZ (4.575s), while ThreeIndex hits the 10s time limit on all 10 runs at n = 40 (Gap = 0.1115) due to O(K n^2) binary variables.

5. Conclusion and References

We have formulated, proved the polyhedral hierarchy of, and experimentally benchmarked five compact MILP formulations and a 2-Index Rounded Capacity Cut Branch-and-Cut algorithm for the CVRP in Julia. Both theory and 10-run benchmarks demonstrate that formulation tightness governs Branch-and-Bound scalability: LiftedMTZ and Single-Commodity Flow (SCF) dramatically outperform basic MTZ and 3-Index formulations on instances up to 40 clients, while dynamic separation of Rounded Capacity Cuts (RCC_BNC) yields the smallest search trees.

- **[1]** M. Desrochers and G. Laporte (1991). Improvements and extensions to the Miller-Tucker-Zemlin subtour elimination constraints. Operations Research Letters, 10(1):27-36.
- **[2]** B. Gavish and S. C. Graves (1978). The Travelling Salesman Problem and related problems. Working Paper OR-078-78, MIT Operations Research Center.
- **[3]** I. Kara, G. Laporte, and T. Bektas (2004). A note on the lifted Miller-Tucker-Zemlin subtour elimination constraints for the capacitated vehicle routing problem. European Journal of Operational Research, 158(3):793-795.
- **[4]** P. Toth and D. Vigo, eds. (2002). The Vehicle Routing Problem. SIAM Monographs on Discrete Mathematics and Applications.
- **[5]** L. A. Wolsey (2014). Vehicle Routing and MIP. 5th Porto Meeting on Mathematics for Industry (Course Lecture 6 reference).